

J2OS Beginner Infrastructure Training Document for Users
Prepared by: Java School Development and Research Team (J2OS)
Date: February 24, 2025
Audience: Enterprise users, managers, and operational teams

Introduction

Welcome to the J2OS infrastructure training! This system is a powerful and simple tool for managing and developing enterprise software such as ERP. With J2OS, you can manage data, configure permissions, execute processes, and generate reports without needing advanced programming knowledge. This document will teach you everything you need to get started and use the system effectively.

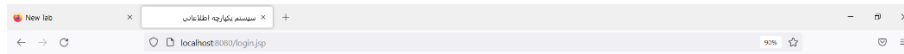
Objectives of This Training

- Logging into the system and getting familiar with the user interface.
- Managing users, groups, and permissions.
- Working with data, processes, and reports.
- Addressing common issues in a simple way.

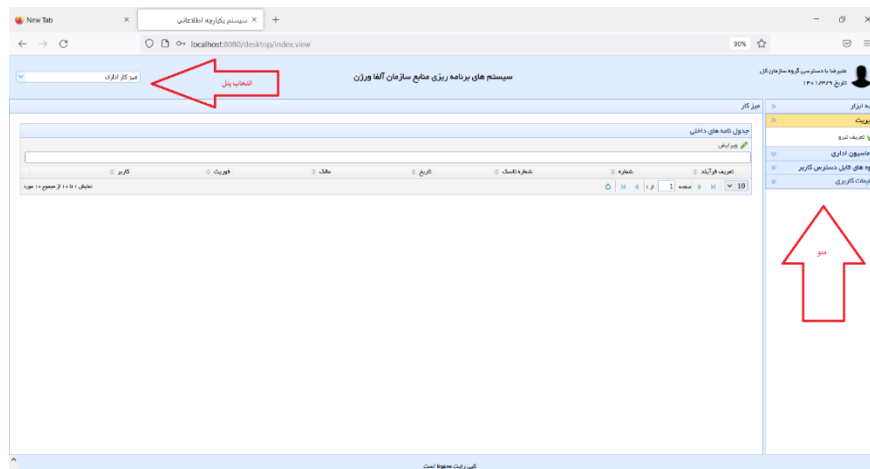
Section 1: Getting Started with J2OS

1. Logging into the System

- Obtaining Login Information:
 - Get your username and password from the system administrator or project coordinator, e.g., username: admin, password: myjava123.
 - System address: http://localhost:8080. If the address has changed, ask the administrator.
- Login Process:
 - Enter the address in your browser.
 - Input your username and password, then click the "Login" button.



- Main Page:
 - After logging in, you'll see the main menu with options like "Panel" and "Administrative Dashboard."

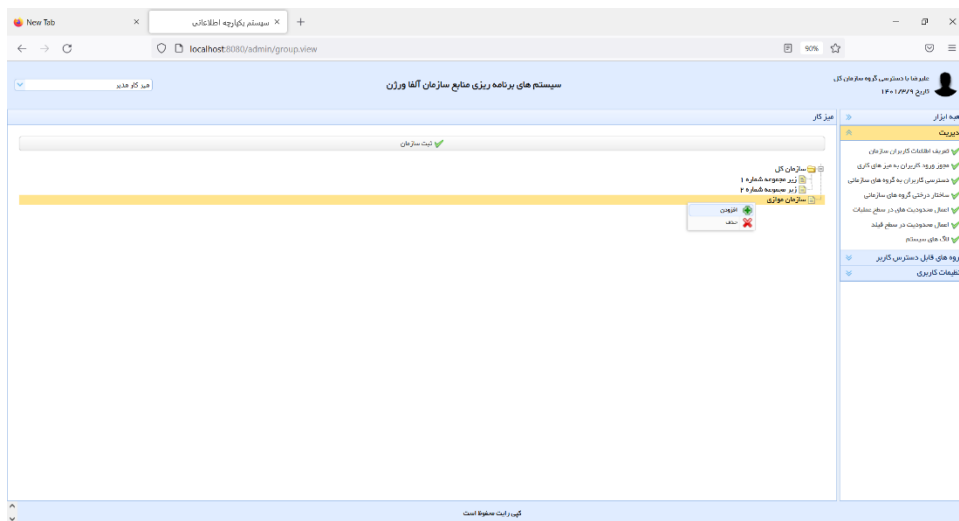


Section 2: Working with the Admin Panel

2. Defining Organizational Structure

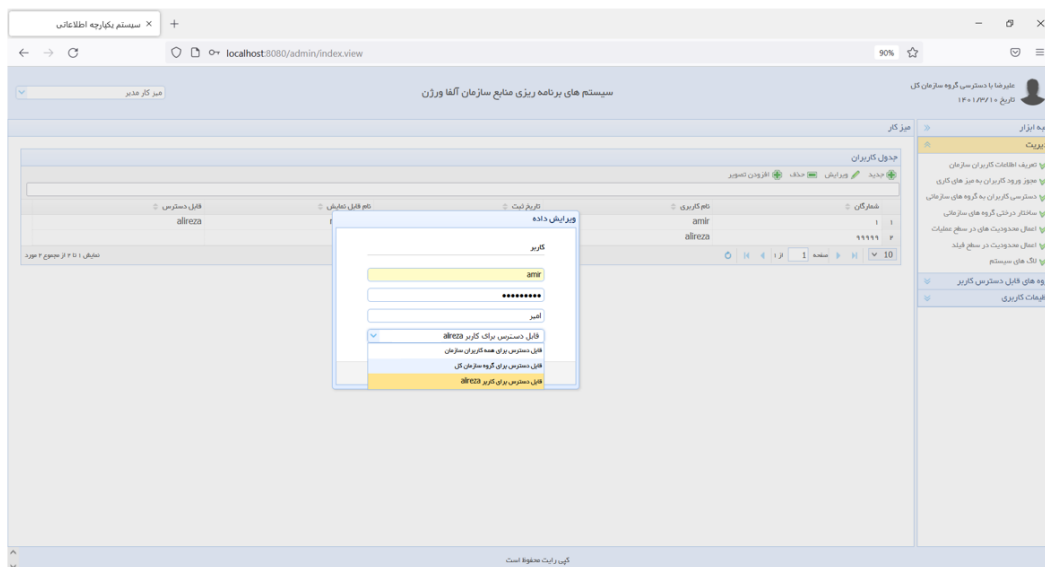
- Accessing the Admin Panel:
 - From the "Panel" menu, select "Admin Dashboard."
- Creating Organizational Groups:
 - Go to the "Organizational Group Tree Structure" section.
 - Click the "Register Organization" button and enter the organization name (e.g., "Sample Company").

- Right-click on "Sample Company" and select "Add" to create subgroups (e.g., "Finance Department" and "Sales Department").



3. Defining Users

- Creating a New User:
 - Go to the "Define Organizational Users" section.
 - Click "New" and enter the details:
 - Name: "Ali"
 - Group: "Finance Department"
 - Confidentiality Level: "Level 2" (visible only to their group).
 - Click "Save."

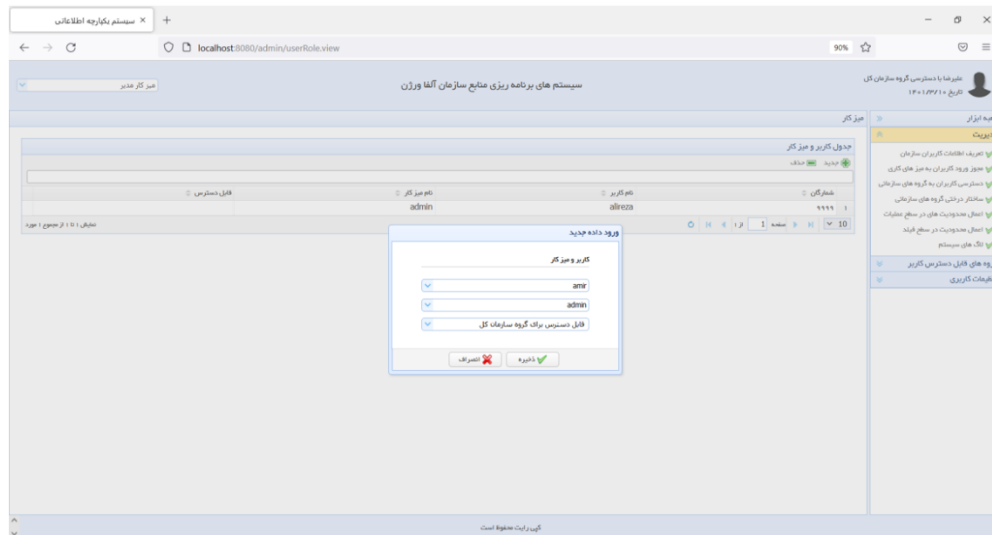


- Confidentiality Levels:

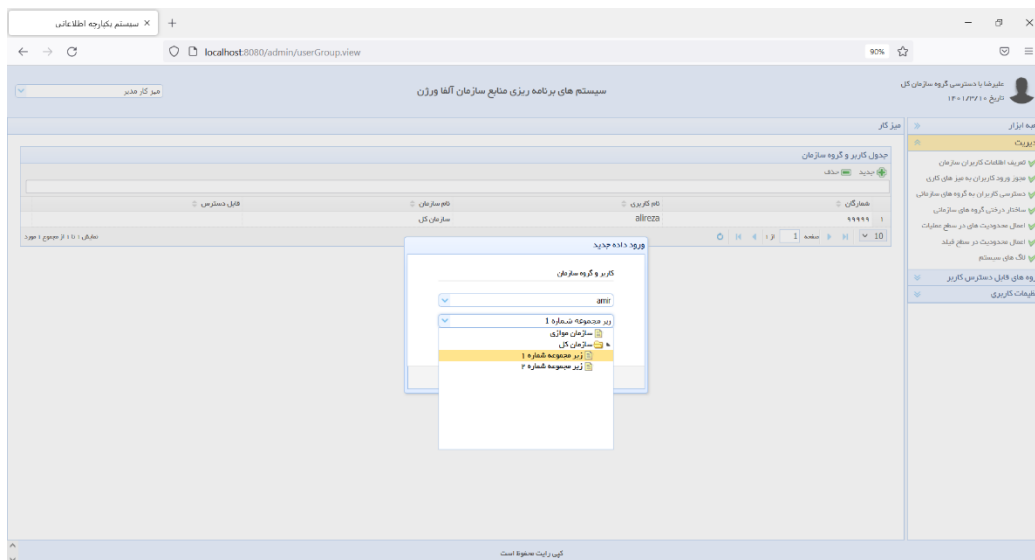
- Level 1: Visible to everyone.
- Level 2: Visible only to the user's group.
- Level 3: Visible only to the user themselves.

4. Configuring Permissions

- Access to Dashboards:
 - Go to the "User Dashboard Permissions" menu.
 - Edit "Ali" and grant them access to the "Admin Panel."

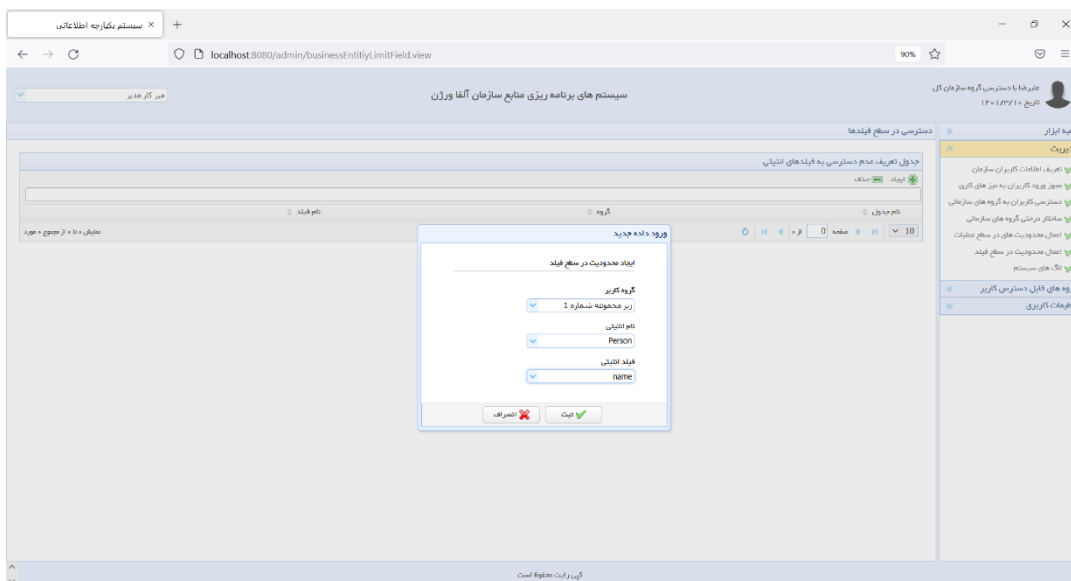
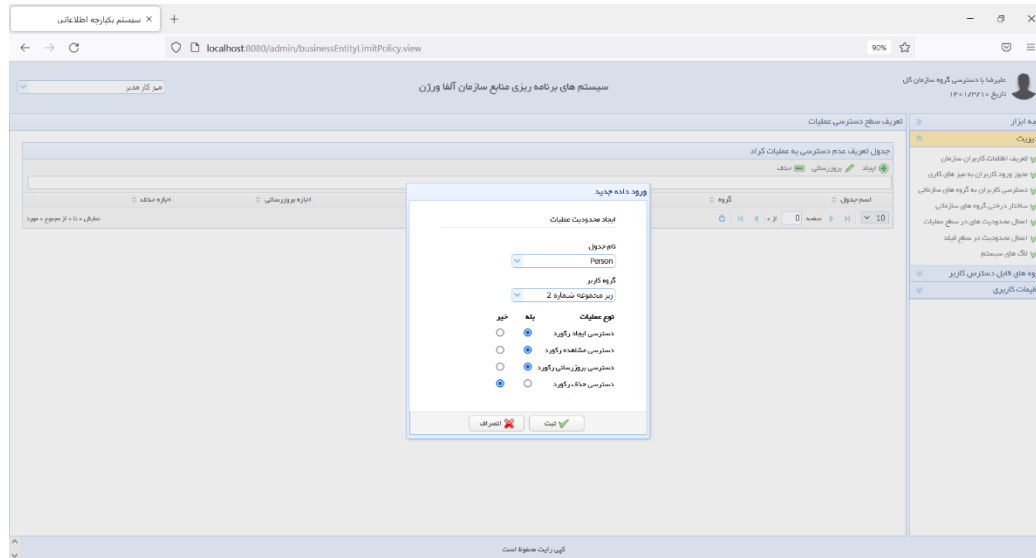


- Access to Groups:
 - Go to the "User Access to Organizational Groups" menu.
 - For "Ali," select the "Finance Department" group and save.



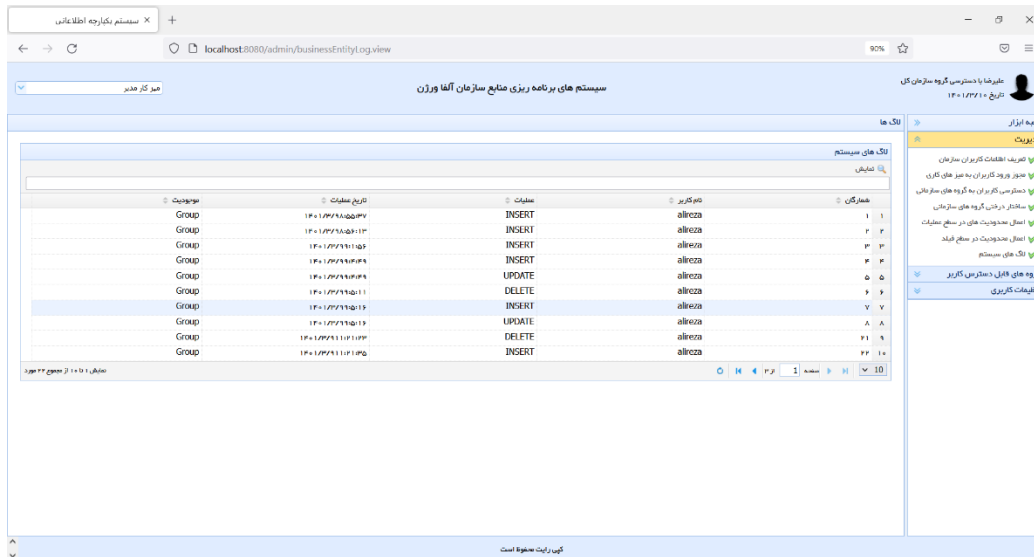
- Restrictions:

- Operations: In "Apply Restrictions at Operation Level," disable "Delete" access to the "Employees" table for the "Sales Department."
- Fields: In "Apply Restrictions at Field Level," disable access to the "Name" field for the "Finance Department."



5. Reviewing Logs

- Go to the "Logs" section.
- View activities such as logins, edits, and views.



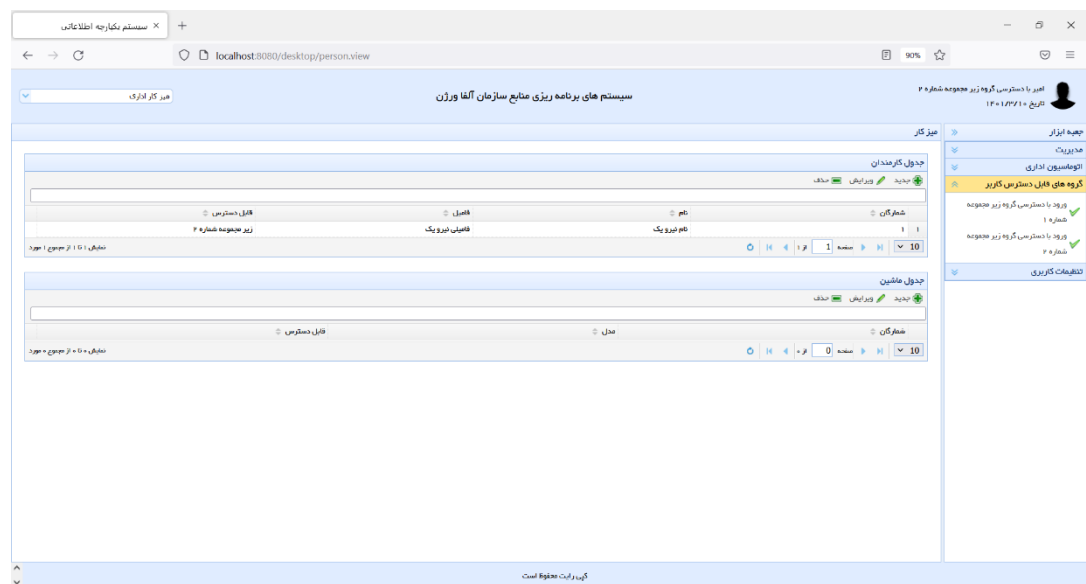
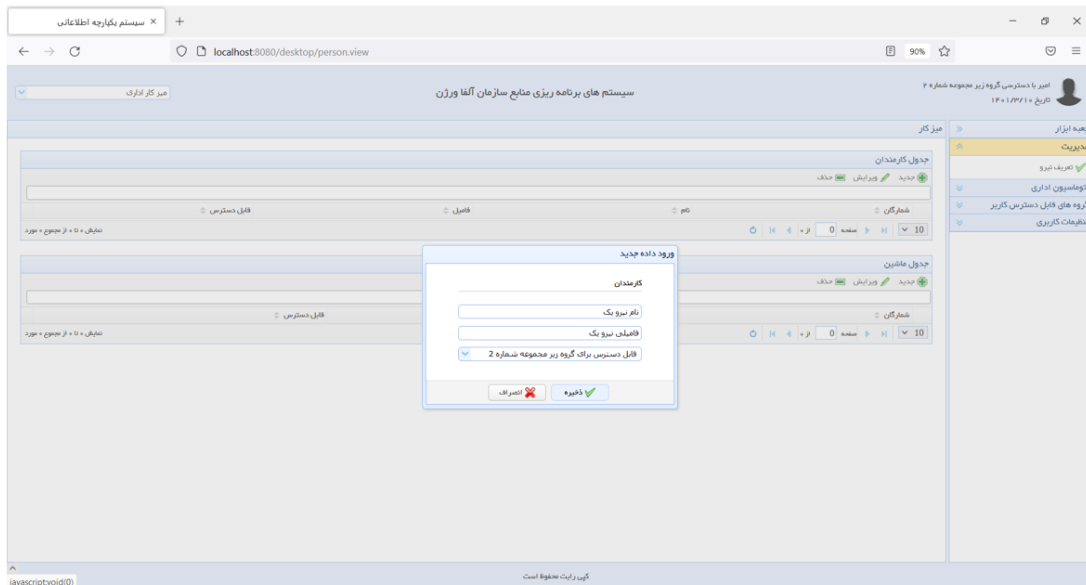
Simple Exercise:

- Add a user "Reza" to the "Sales Department" and disable their "Edit" access. Then check their login log.

Section 3: Working with the Administrative Dashboard

6. Adding and Managing Data

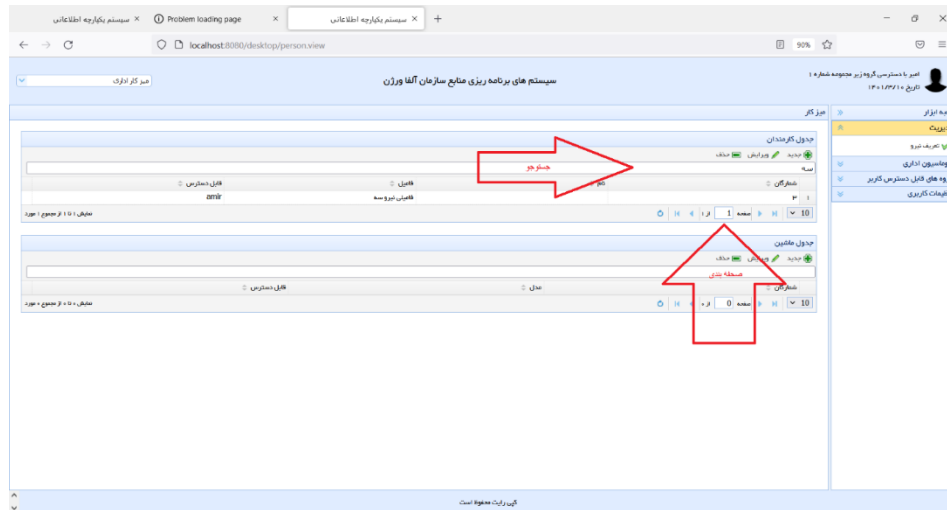
- Accessing the Administrative Dashboard:
 - From the "Panel" menu, select "Administrative Dashboard."
- Creating an Employee:
 - Go to the "Define Personnel" section.
 - Enter details (e.g., Name: "Person 1," Group: "Finance Department").
 - Set the access level to "Finance Department" and save.



- Adding a Vehicle:
 - Click on "Person 1," select "Add Vehicle," and enter vehicle details (e.g., "Pride").

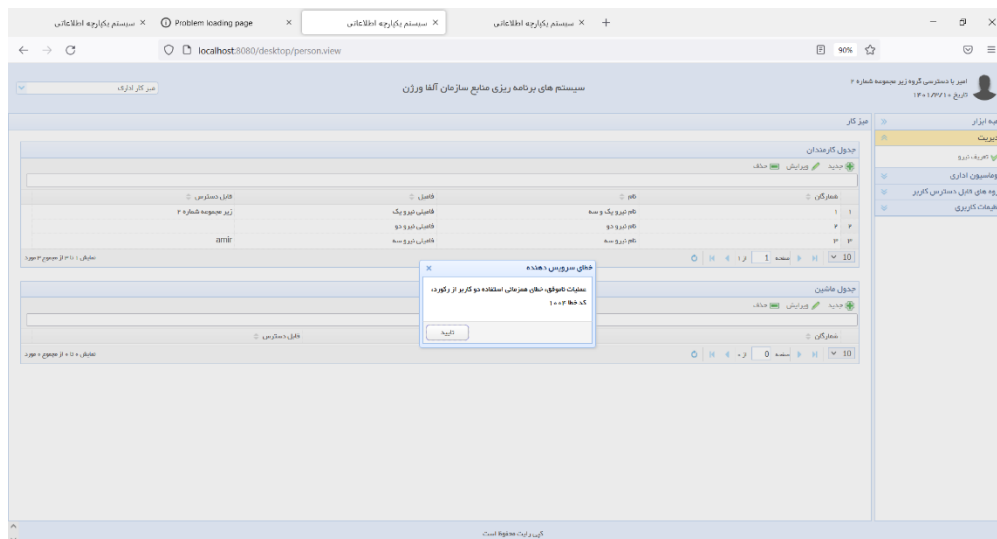
7. Searching and Viewing

- At the top of the "Employees" table, type "Person 1" in the search box.
- See pagination at the bottom of the table and select page two.
- Note: If your group doesn't have access to the "Name" field, you won't see it.



8. Resolving Conflicts

- If two people edit an employee simultaneously:
 - The first person saves their changes.
 - The second person sees a "Record Already Modified" error.



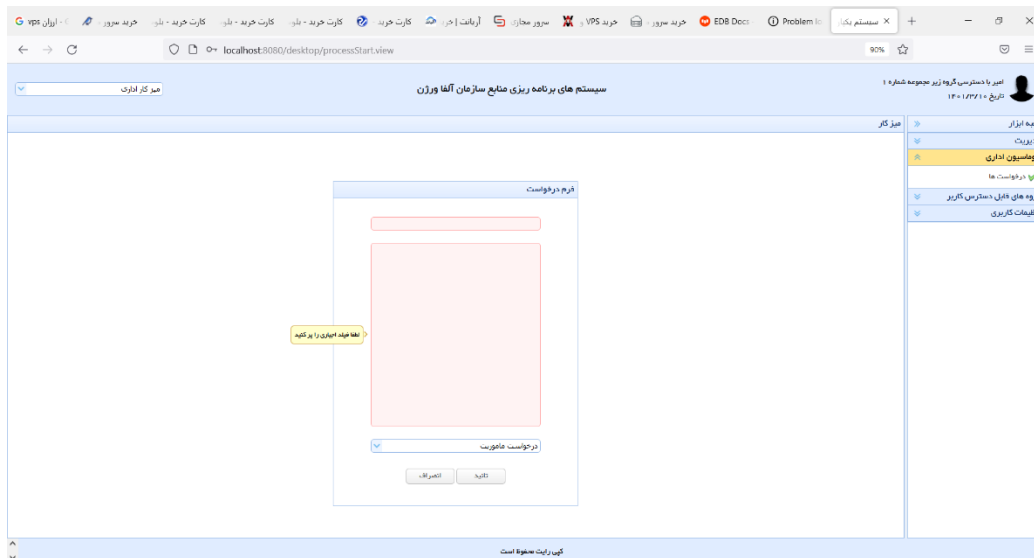
Simple Exercise:

- Add two more employees:
 - "Person 2" (Access: Entire organization).
 - "Person 3" (Access: Only themselves).
- Change the group and see which employees are displayed.

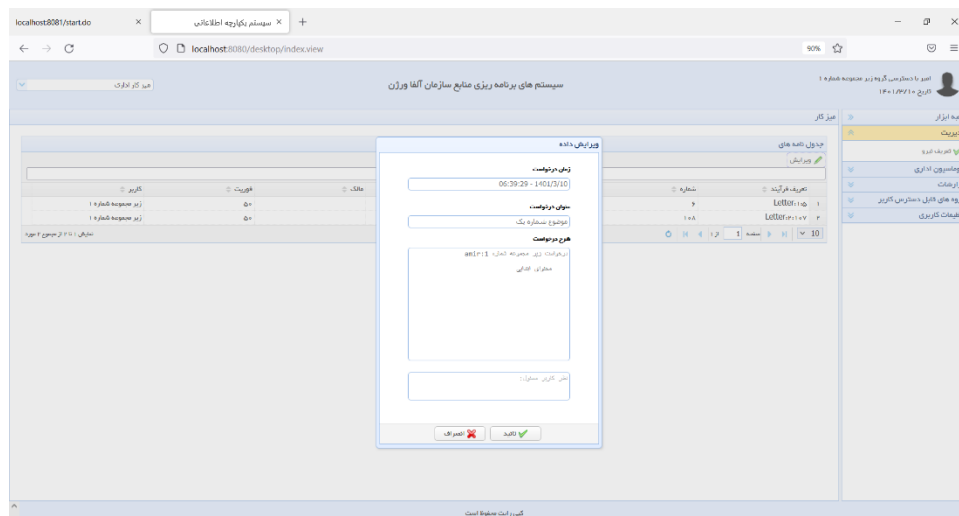
Section 4: Working with Processes

9. Executing a Process

- Accessing Automation:
 - Use the "Administrative Automation" menu.
- Starting a Process:
 - Select the "Leave" process.
 - Enter a subject (e.g., "Monday Leave") and text, then click "Confirm."
 - The process moves from "Finance Department" to "Sales Department," and an SMS is sent.



- Tracking:
 - View the process in your inbox (e.g., "Finance Department").
 - Add your comment (e.g., "Approved") and click "Confirm."



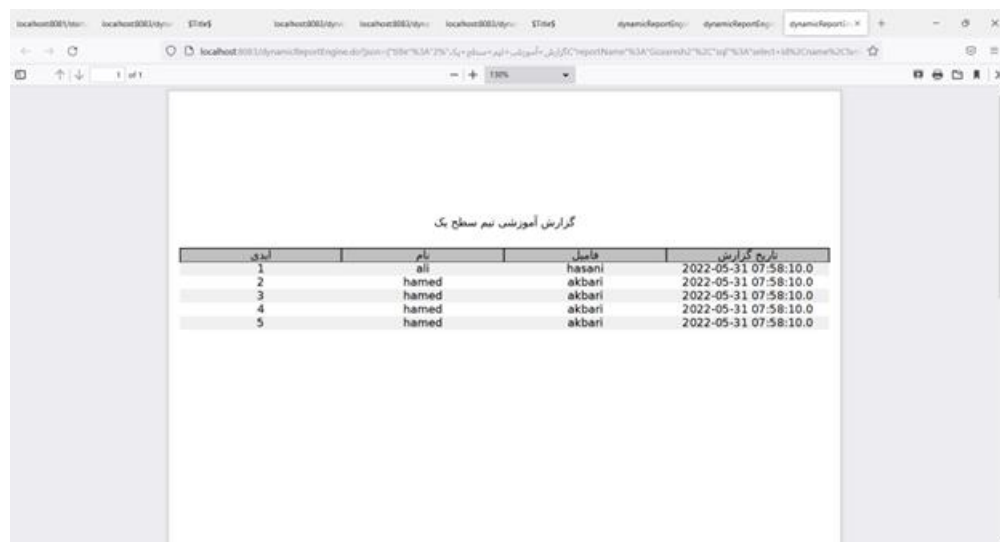
Simple Exercise:

- Start a "Leave" process and advance it to the "Sales Department" stage.

Section 5: Reporting

10. Creating Reports

- Accessing Reports:
 - Use the "Reports" menu.
- Three Report Types:
 - Type 1: Predefined report with SQL, e.g., Employee List.
 - Type 2: Predefined report without SQL, data sourced from the system.
 - Type 3: Automatic report with simple SQL, e.g., SELECT * FROM Person.
- Execution:
 - Create an automatic report with the above SQL and open the output link.



ردیف	نام	فامیل	تاریخ گزارش
1	ali	hasani	2022-05-31 07:58:10.0
2	hamed	akbari	2022-05-31 07:58:10.0
3	hamed	akbari	2022-05-31 07:58:10.0
4	hamed	akbari	2022-05-31 07:58:10.0
5	hamed	akbari	2022-05-31 07:58:10.0

Simple Exercise:

- Create a Type 3 report with custom SQL (e.g., list of vehicles) and view the output.

Section 6: Practical Tips and Support

11. Key Tips

- Changing Groups: Use the "User Accessible Groups" menu to switch your active group.
- Logout: Always use the "Log Out" option from the settings menu.

- Issues: If something doesn't work, check the logs or contact support.

Section 7: Module Development

Why Low-Code?

J2OS eliminates the need for complex coding with automated tools like Crud Repository and built-in security. Defining an Entity:

@Entity

```
public class Person {  
    @Id  
    private Long id;  
    private String name;  
    @ManyToMany  
    private List<Car> cars;  
}
```

@Entity

```
public class Car {  
    @Id  
    private Long id;  
    private String model;  
}
```

Explanation: A Many-to-Many relationship is defined between employees and vehicles. Using Crud Repository

@Service

```
public class PersonService {  
    @Autowired  
    private CrudRepository<Person, Long> personRepo;
```

```

public Person findPerson(Long id) {
    return personRepo.findOneByIdOnPermission(id);
}

public List<Person> search(String name) {
    Map<String, Object> params = new HashMap<>();
    params.put("name", name);
    return personRepo.findAll(params, "name");
}
}

```

Feature: Search, pagination, and permissions are applied automatically.

Building a REST API

```

@RestController
@RequestMapping("/api/persons")
public class PersonController {
    @Autowired
    private PersonService service;

    @GetMapping("/{id}")
    public Person getPerson(@PathVariable Long id) {
        return service.findPerson(id);
    }
}

```

User Interface

- Use the J2OS JS library for UI without needing Angular.

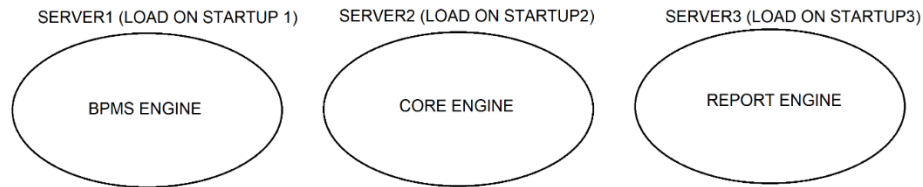
```
<j2os-table entity="Person" fields="id,name"></j2os-table>
```

Explanation: Generates an employee table with minimal code.

Comprehensive Exercise:

- Create an "Order" entity with fields "id" and "amount."
- Write a service to search orders and an API to display them.
- Build a UI table for "Order."

Section 8: Deployment and Maintenance



Tools

- Database: Oracle or PostgreSQL
- Server: Tomcat or WebLogic
- Frameworks: Spring, Hibernate, Camunda, JasperReport

Deployment Steps

- Install the database and server.
- Upload J2OS files.
- Configure NGINX for distribution.

Maintenance

- Regularly check logs.
- Apply updates from J2OS.

Section 9: Tips and Support

Practical Tips

- Change Group: Use "Accessible Groups" to switch.
- Troubleshooting: If something fails, check logs or contact support.
 - Support Email: **info@j2os.org**